

Google Capstone - Cyclistic
2025 / 2nd Quarter/ April

Navigating Cyclistic to Speedy Success

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Business task: Determine how our different customer types use cyclistic services differently.

We will address:

- Cyclistic members
- Cyclistic casual users



Project Roadmap



Data sources

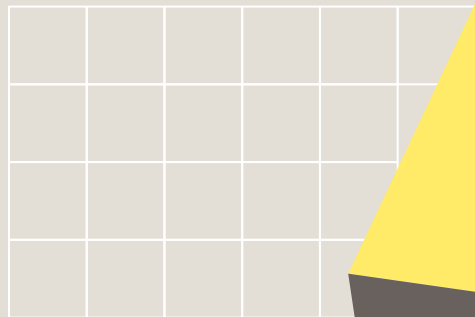
Cyclistic data source

All cyclistic data is presented here:

<https://divvy-tripdata.s3.amazonaws.com/index.html>

All 12 months of 2025 cyclistic data
will be used for the analysis

Step 2



Processing Data

A python Jupyter Notebook was created to store all monthly data in it's own dataframe. The python Padanas and Numpy engine were used to help perform data manipulations and sort the dataframes.

Data processing reveals that there are four data types that are needed for this analysis. They are:

1. Rideable Type - Mechanical or Electrical Bike
2. Started At - Time stamp of ride start time
3. Ended At - Time stamp of end of ride time
4. Member/Casual - Member or Casual Customer

Step 2

Ensuring Data Consistency

In order to perform an accurate analysis, the Jupyter Notebook discussed on slide 6 is used to ensure data consistency across the 4 required categories. Steps Taken include:

1. Ensure rideable type entries were limited to 'classic_bike' or 'electric_bike'
2. Ensure member_casual entries were limited to 'member' or 'casual'
3. Ensure that the ended time stamp was later than the started time stamp
 - a. 29 entries were found for december to violate this condition. Since the dataset was large, these entries were simply removed

Step 3

Analysis:

Once the data was determined to be clean enough for analysis, the Jupyter Notebook was used to analyze the data. In order to do this, in combination with the Pandas and Numpy packages, python commands were used to create two pivot tables, described below.

- The first pivot table was used to describe how member vs casual renters use the bikes on a monthly basis
- The second pivot table was used to describe how long members vs casual riders rent the bikes on a weekly basis

Both pivot tables are shown in the following slides

Company Name

Pivot Table: Monthly Data

month	total_rides	mem_elec_bike	mem_classic_bike	casual_elec_bike	casual_classic_bike	member	casual	avg_mem_duration_mins	avg_casual_duration_mins
January	178372	82616	57380	25228	13148	139996	38376	10.76379232263779	17.755763498019594
February	138689	71553	42983	17518	6635	114536	24153	10.073934920025145	14.710406988779862
March	151880	78777	45349	20347	7407	124126	27754	9.993693102170374	14.397951646609497
April	298155	132715	79571	54427	31442	212286	85869	11.267538273838124	20.85386821786675
May	371341	163157	98968	69552	39664	262125	109216	11.326354563662374	21.507462642836217
June	502456	201058	118761	115302	67335	319819	182637	12.115591068698231	23.837889912777808
July	678904	242980	143831	181809	110284	386811	292093	12.952918350305449	25.558535843036292
August	763432	284399	155696	213782	109555	440095	323337	13.351122825753531	24.340118235772582
September	790177	289545	162894	219651	118087	452439	337738	13.099092010193639	23.52954971013034
October	714759	291264	158034	184166	81295	449298	265461	12.674150252171163	21.779956377773008
November	646039	278726	143303	157385	66625	422029	224010	12.369323695764981	20.459260434802015
December	356599	169313	88146	71836	27304	257459	99140	12.152375718075499	18.23898053258019

Year/Quarter/Month

Company Name

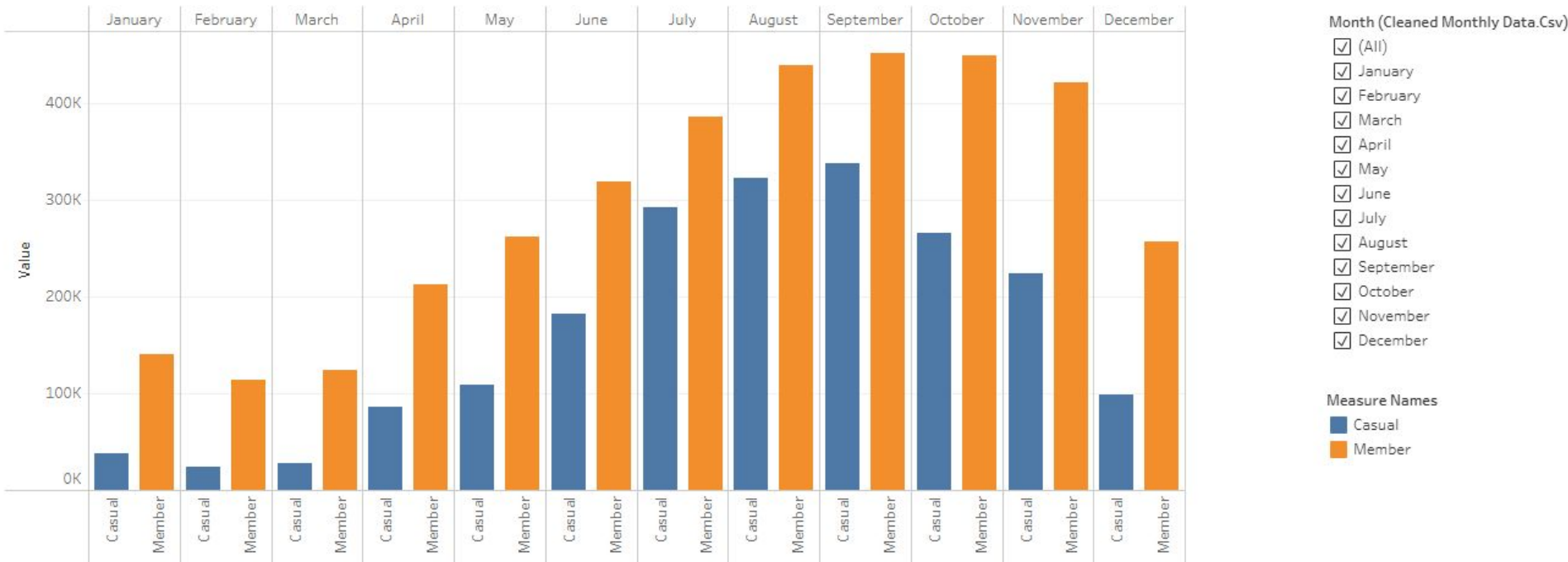
Pivot Table: Ride Duration Per Day

day	member_casual	jan_avg	feb_avg	mar_avg	apr_avg	may_avg	jun_avg	july_avg	aug_avg	sep_avg	oct_avg	nov_avg	dec_avg
Sunday	member	11.91	10.53	10.49	12.17	12.78	13.1	14.01	14.77	14.69	13.88	14.28	12.61
Sunday	casual	23.01	13.0	17.47	23.59	26.08	28.04	27.6	26.46	25.85	27.44	25.73	21.76
Monday	member	10.86	9.19	9.84	10.78	11.29	11.7	12.3	13.12	12.11	12.57	12.23	11.19
Monday	casual	17.65	13.42	13.88	18.02	23.34	24.01	24.37	24.55	21.93	23.23	20.35	16.65
Tuesday	member	10.41	10.45	9.9	10.71	10.89	11.67	12.67	13.01	12.15	12.32	11.73	11.37
Tuesday	casual	13.4	15.58	13.53	17.96	19.41	21.23	22.8	22.25	20.23	18.41	17.79	14.24
Wednesday	member	10.3	9.87	9.43	10.24	10.83	11.25	12.29	12.74	12.76	12.17	11.77	12.29
Wednesday	casual	14.52	15.55	11.62	13.32	16.33	19.38	22.1	21.85	20.71	17.03	15.88	13.79
Thursday	member	10.37	10.1	10.01	10.7	10.84	11.78	12.5	12.43	12.85	12.39	11.91	12.04
Thursday	casual	16.97	14.03	14.23	16.74	18.25	20.77	23.08	21.11	22.73	18.29	17.18	14.42
Friday	member	10.58	10.04	10.06	12.22	11.27	12.27	12.67	13.72	12.71	12.45	12.07	12.46
Friday	casual	16.23	12.89	12.89	23.32	21.09	23.76	24.67	26.17	22.64	21.45	19.37	20.37
Saturday	member	11.21	10.6	10.61	12.24	12.28	13.35	14.51	14.82	14.28	13.56	13.63	13.22
Saturday	casual	20.6	17.71	17.36	24.39	23.96	26.12	29.39	26.95	25.43	23.23	23.36	20.93

Year/Quarter/Month

The graphic below shows an important inference based on the monthly pivot table: Monthly sales are always dominated by members as opposed to casual users.

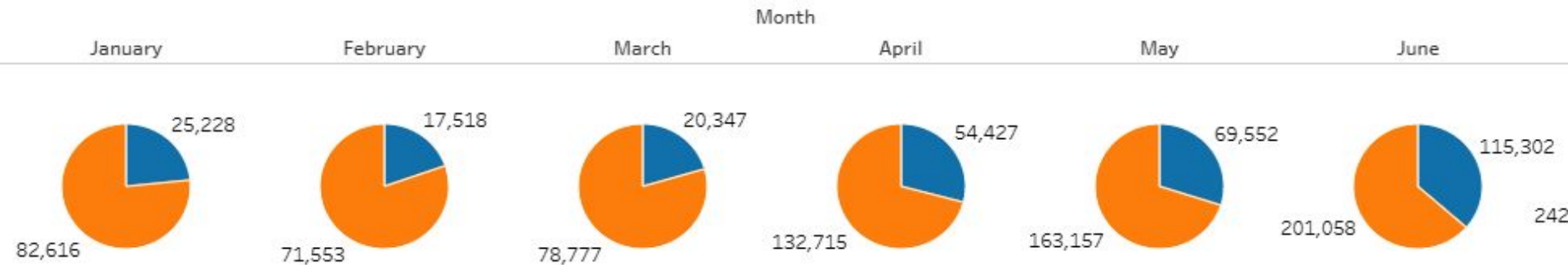
Member vs Casual Rental per Month



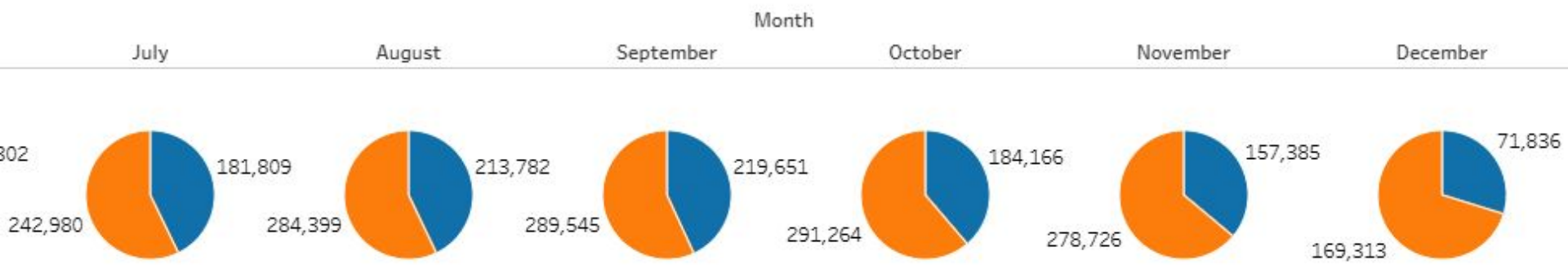
Company Name

In addition, it can be shown that members dominate bike rentals, especially the shown electrical bikes. This happens most predominantly in cold months such as January to March as well as November to December. During the warmer spring to autumn months, casual rentals are almost equal member rentals.

Electric Bike Rentals Members vs Casual Users



Electric Bike Rentals Members vs Casual Users

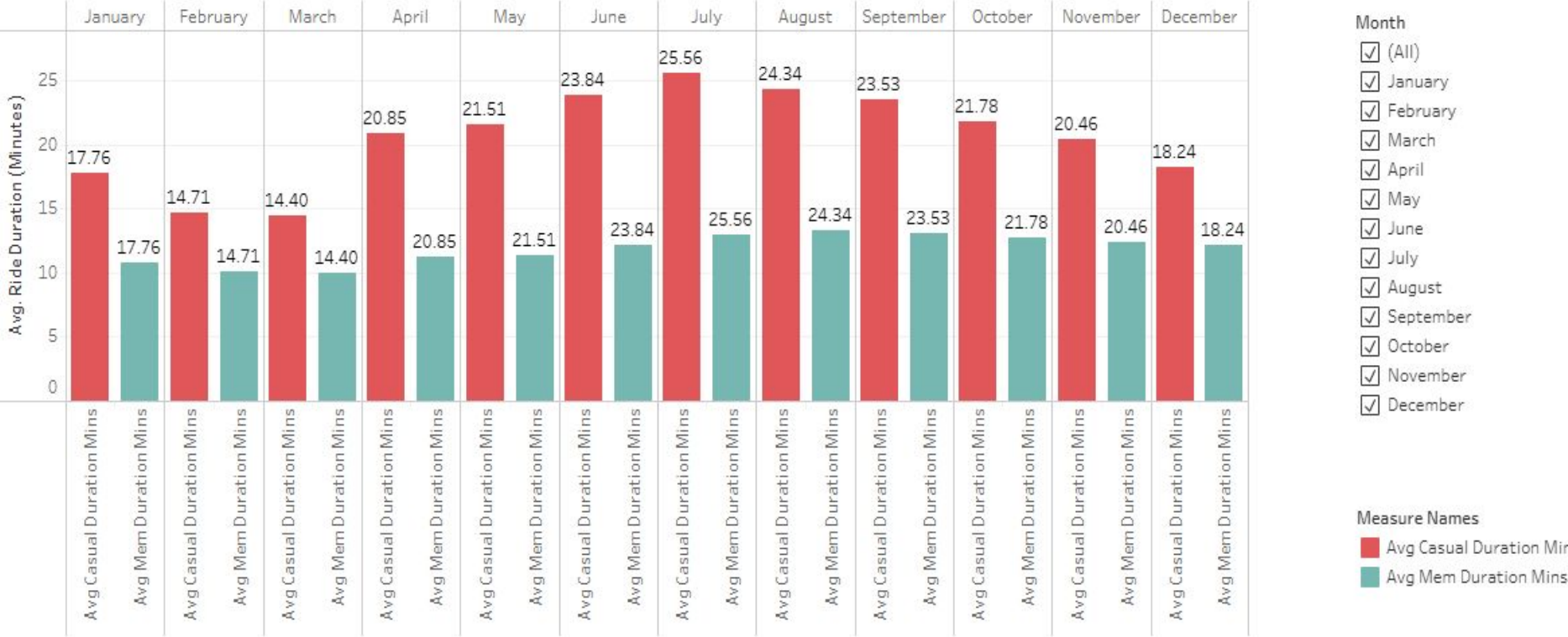


Analyzing the above pivot tables also shows a really interesting trend. Casual riders tend to rent the bike for a much longer period than the average member rents them for. The data is broken up into per/month and per/day of the week to demonstrate this trend on the next two slides.



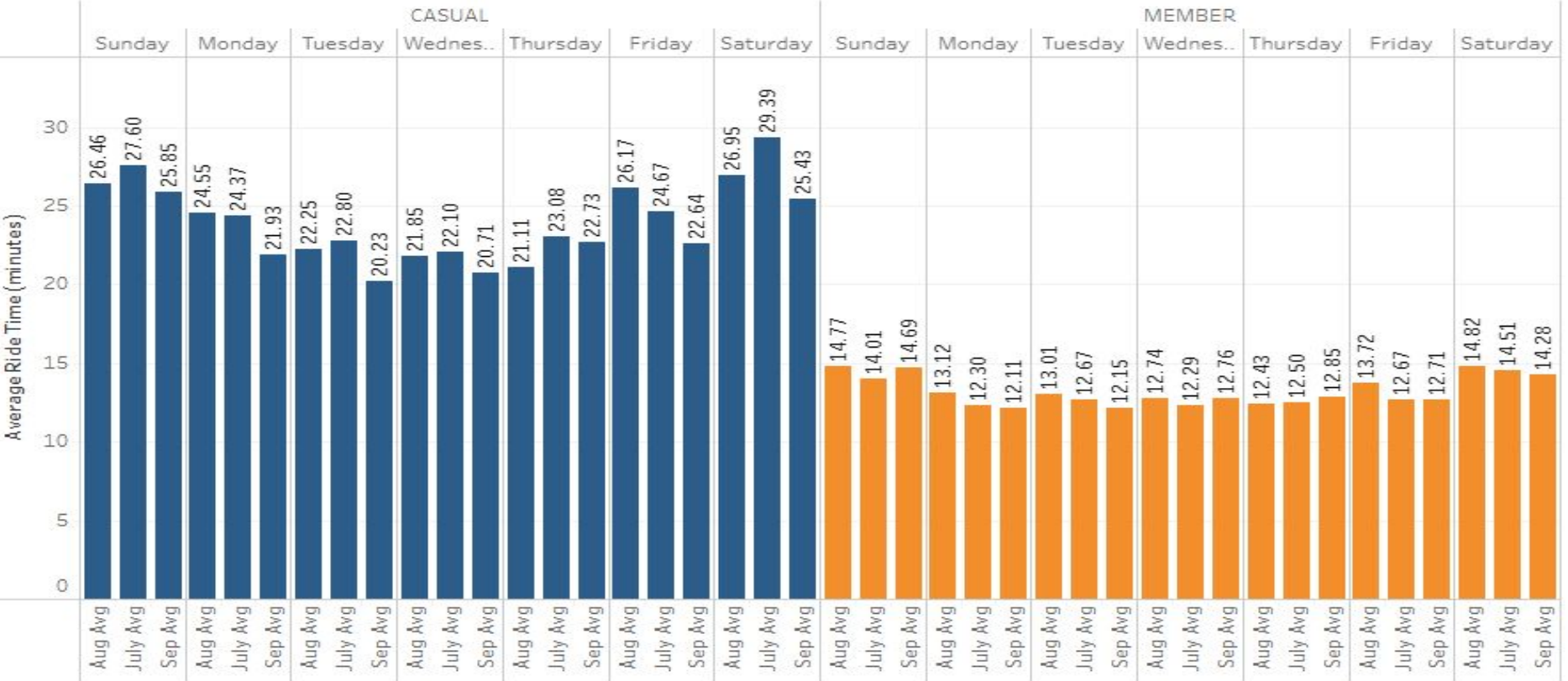
Shown below is the average rental time plot for casual vs member renters. The plot shows a clear trend of casual riders renting the bikes for far more time than the member renters. This trend becomes especially pronounced during the warm weather months of April to September.

Average Ride Time (minutes) Member vs. Casual Monthly



Shown below is the average ride time for members vs casual riders, on a daily basis. There are three months specifically selected for presentation: July, August, and September. These three months are chosen for presentation because they represent the closest to 50/50 member to casual number of rentals.

Average Ride Time (minutes) Member vs. Casual Weekly

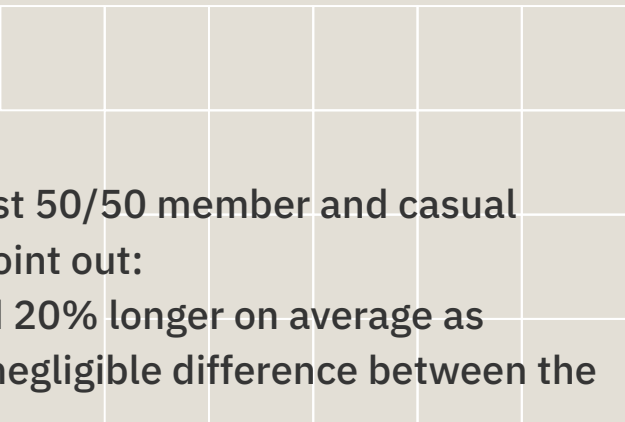


Initial Findings Outlook

Reviewing the data in relation to the business question, “how do members and casual riders use the cyclistic service differently?” provides the following findings:

1. Members clearly rent more times per month than casual riders.
2. Casual riders rent less times per month than member, but on average, they rent the bikes for a longer period of time.
3. Based on the data, seasons seem to play a strong role in how cyclist use the service:
 - a. During colder winter months, members tend to rent the electric bike, presumably to get around faster. In addition, during these months, casual rentals is significantly lower than other months.
 - b. During warmer month, such as spring, summer, or some of autum, casual ridership increases dramatically, almost to the same levels as member ridership, and the average ride time is much greater.

The Bottom Line



So why is this relevant?

1. Even though during the warmer months, where ridership is almost 50/50 member and casual renters, there is a big catch that analyzing weekday trends will point out:
 - a. For the casual rider, ride times on weekend days are around 20% longer on average as opposed to week days. Members on the other hand have a negligible difference between the weekend days and weekends.

Therefore:

Since members use the bikes much more in cold weather, use the electric bikes more during these months (presumably to get around quicker), and constantly use the bikes for the same times during the week, members are probably using cyclistic as a reliable means of daily transportation.

On the other hand, casual riders are using the bikes during more warm weather months for longer periods, and are probably using the bikes for hobby riding and convenient transportation to events on weekends.

In order to visual the data in this project more effectively, the following tableau dashboards were created:

https://public.tableau.com/views/Google_CapStone/Dashboard3?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

[https://public.tableau.com/views/Google_CapStone/Dashboard4?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share link](https://public.tableau.com/views/Google_CapStone/Dashboard4?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)



Reccomendations

Based on the analytics, here are several recommendations:

1. Market to casual riders the benefits of replacing their daily transportation with a cyclistic option.
2. Expand the availability of cyclistic classic bikes during the warmer months and electric bikes during the cooler months.
3. Market to members different excursions that they might be interested in that they could take using a cyclisitic bike, to increase member rental times.



Questions?